



An Approach to Inter-annotator Agreement Evaluation for the Named Entities Annotation Task at OpenCorpora

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Abstract. In this paper we introduce an approach towards inter-annotator agreement (IAA) evaluation for tasks with an unrestricted set of possible choices. IAA is an important tool because it provides direct feedback to both annotators and annotation guidelines developers. There are several approaches towards IAA evaluation for tasks with a closed set of alternatives like morphological annotation; adaptation of these to tasks with an unrestricted set of possible choices is not fully applicable. An *a posteriori* evaluation approach is introduced in this paper in order to evaluate current inter-annotator agreement for crowdsourcing on the stage before moderation. The proposed method is based on Venn diagrams naturally representing opinions of multiple annotators selecting an arbitrary number of word sequences. Four categories of agreement are pointed out, and the contribution of each is considered and visualized along with proportions of their investment. One total criterion is selected to evaluate the agreement. The approbation results are discussed for evaluating data of named entities annotation task in the OpenCorpora project in Russian. The recommendations are given on the approach applicability for evaluating IAA in unmoderated named entities annotation data with open set of alternatives.

Keywords: Inter-annotator agreement · Named entities annotation · Named entities recognition · Crowdsourcing · Corpora · Annotation with open set of alternatives

1 Introduction

Manually annotated corpora are of big importance in most natural language processing tasks. One particular application is named entities recognition. A training set was required for named entity recognition machines training and evaluation in the context of

The reported study was supported by RSF research project 19-18-00547.

the FactRuEval information extraction competition held as part of the Dialogue Evaluation initiative in the run-up to Dialogue 2016 [15]. The OpenCorpora project attempted at collecting a satisfactory amount of such data via crowdsourcing. A conceptual model was introduced in [11] along with guidelines for both annotators and contestants. The annotations guidelines were intended for Russian native speakers and didn't require any prior linguistic knowledge from volunteers. These guidelines described entities types and their interpretation rules for several markup layers [13]. The dataset annotated was to be verified by a moderator, but automated techniques of inter-annotator agreement (IAA) evaluation are required on early stages of annotation [4], namely before any moderation.

The aim of this research is developing an approach to automated IAA evaluation for the named entities annotation (NEA) task, which is based on selecting chains of words via entities recognition by annotators (as opposed to choosing from a closed set of alternatives). Here the alternatives variety is not limited, but by a particular paragraph size. Moreover, entities might be absent or intersecting each other.

Discrepancy between annotators is a natural phenomenon which results from differences in skills, knowledge and experience, as well as from tiredness or data ambiguity [16]. Such discrepancy is subject to verification by a moderator on later stages of annotation, but the setting of NEA task should involve sharing the same understanding by annotators. In supposition that all of the volunteers are conscientious, the more similar the independently created precedents of markup of a paragraph are, the more successful the markup setting is. This research is dedicated to evaluating resemblance of a set of independent opinions, expressed by volunteers during NEA.

1.1 Named Entities Annotation in OpenCorpora

The conceptual model of NEA in OpenCorpora [11] consists of following sub-layers:

- (1) words spans (corresponding to entities attributes);
- (2) objects, or mentions (object attributes grouped for each mention);
- (3) coreference between objects;
- (4) identification with an external database (namely WikiData).

Raw data consist of paragraphs which are subject to annotate. Each paragraph is annotated by 4 independent volunteers who select entities' attributes (further referred to as spans) basing on NEA markup guidelines. A span is an uninterrupted chain of tokens, marked with one of the predefined tags. Each volunteer is given an interface [12] allowing markup of an arbitrary number of spans, which can intersect each other or even share borders, see Fig. 1. This is the first layer of the NEA markup.

Spans represent entities attributes, namely 9 tags describing person (surname, name, patronymic, nickname), occupation (job), organization (name and descriptor) and location (name and descriptor) attributes. Grouping spans results in annotating mentions on the second layer of markup. E.g., one can annotate two organizations and two locations out of spans represented on Fig. 1. Several spans on Fig. 1 are nested; consequent spans can share borders as well. An example of further identification of entities can be found in [15].

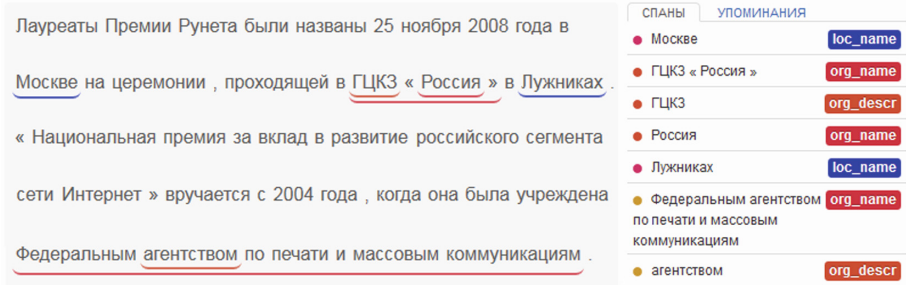


Fig. 1. Spans (entities' attributes) annotation in the OpenCorpora interface

This work is concerned with evaluation of the first layer markup, i.e. spans, each represented by its length in tokens, index of its first token, a tag, the index of comprising paragraph and the ID of the volunteer who annotated it. A match of two spans is defined as match of borders, tags and comprising paragraphs.

1.2 The Purpose of Measuring the IAA for the NEA Task

To achieve acceptable quality, the annotation is conducted by volunteers with 4-fold overlap, and the final decision is taken by a moderator. Moderators have proven skills and usually have a linguistic education. In this workflow, the moderation is often a bottleneck as it can happen in several months or even years after all volunteers have finished their particular piece of work. The absence of feedback is a demotivating factor for volunteers. Early feedback is also important for guidelines developers because it can help to improve guidelines over time. IAA can be a marker serving for early feedback.

The purpose of this paper is to introduce an approach to count and evaluate IAA in a particular setting of the NEA crowdsourcing task, as this task is not that of picking one of a few listed alternatives, but it was set as recognizing and selecting tokens and chains of tokens (zero to any necessary amount) out of a paragraph. E.g., Fig. 1 shows a sample (in Russian) where an organization entity “ГЦКЗ «Россия»” is annotated with 3 spans: a short organization name “Россия” (Russia), a descriptor “ГЦКЗ” (an abbreviation standing for “Central state concert hall”, which can be written in four tokens as well) and a full organization name consisting of both tokens and quotation marks. Thus complex names can consist of several tokens and/or token chains, which can be selected without any limitation but that of a given paragraph limits.

Two approaches exist towards IAA, which are discussed in Sect. 2. Existing approaches are good at estimating IAA when the choice is limited to few alternatives (as in morphological markup crowdsourcing tasks with fixed number of alternatives: an annotator picks one of 6 cases or one of several tenses for a given word). But in the spotlighted task numeric values for existing criteria would approximate zero as the denominator – the amount of possible alternatives for a paragraph – is enormous and follows from combinatorics. These approaches are stated to be inapplicable here.

In Sect. 3 a different approach is introduced towards IAA evaluation for the NEA task with open set of alternatives (spans to annotate in a paragraph; spans might be intersecting each other). The results of the approbation are discussed in Sect. 4.

2 Overview of Approaches Towards IAA Evaluation

The fundamental supposition for the methods of IAA evaluation is that data are considered reliable if volunteers agree significantly on assigning equal tags to equal spans. If independent volunteers systematically tend to annotate the same subjects similarly, the conclusion is that they share the same understanding of the guidelines and that they act according to them [1]. Two approaches exist towards IAA evaluation: the aposteriori approach and the apriori approach.

The first category is represented with the observed agreement criterion – “the percentage of judgments on which the two analysts agree when coding the same data independently” [14, p. 323]:

$$A_o = \frac{1}{i} \sum_{i \in I} agr_i,$$

$$agr_i = \begin{cases} 1, & \text{if the two annotators assigned item } i \text{ to the same category,} \\ 0, & \text{if the two annotators assigned item } i \text{ to different categories.} \end{cases}$$

The second approach is calculating chance-corrected agreement [6], which is based on the apriori estimation of the probability A_e of agreement expected by chance. Following values are evaluated: agreement over and above chance $1 - A_e$, and observed agreement beyond chance $A_o - A_e$. This approach uses the latter two criteria in various combinations and comprises criteria S [2], Scott’s π [14], Cohen’s κ [5] and their generalizations, along with Krippendorff’s α [10] expressed in terms of disagreement [1]. The S , π and κ criteria are based on different assumptions put in the basis of calculating of the probability that the annotator would assign an arbitrary item to the specified category [7, 17].

The approach concerning the chance agreement probability is most applicable to tasks with a closed set of alternatives to choose from, as in the case of morphological annotation: each word has a closed set of morphological categories to assign [3]. In this setting blind random assignment is possible within the reward-based crowdsourcing. In the latter case random choice probability is implied, which serves base to a probability of agreement by chance.

But this latter approach wasn’t applied in the NEA task, because it’s quite the contrary: the markup process is based on entities recognition in text, thus the ground for assigning categories is subject to annotate as well (moreover, each token can be included in several spans). A volunteer should annotate an arbitrary number of spans (chains of tokens, perhaps, intersecting, as in Fig. 1; a paragraph can contain no spans at all) from a paragraph, and provide each span with a tag stating the type of the attribute annotated, according to the guidelines. A probability of annotating a particular span could be calculated, but it would be extremely small; furthermore, pointing out an apriori empirical value of agreement, achievable by chance, doesn’t seem to be helpful. Even filtering the candidate tokens by part of speech for adjusting the probability of a random selection wouldn’t be a good practice, for locations and organizations names can include propositions (“The house by the river bank”), pronouns, adverbs, verbs (“Darling, I’ll call you back” café). As a result, the aposteriori approach towards IAA for NEA is chosen, including evaluation of intersecting sets of opinions represented as spans.

3 A Procedure for IAA Evaluation for the NEA Case

Each paragraph is examined by 4 volunteers during markup. While the markup is not finalized, the corpus contains paragraphs with fewer opinions expressed regarding it. As the corpus grows, the data is verified by the moderator. Of particular interest is evaluating consensus within groups of volunteers: among 2, 3, 4 annotators. These sets of opinions are a maximum of information available at the moment of evaluation. Pairwise comparison of opinions in groups of three or four volunteers won't be conducted, for that would be only partial data, whilst the focus is on exposing quorum of a full group of volunteers, thus evaluating the aggregate picture.

Four volunteer opinions are considered as valuable in OpenCorpora. In a group of four a quorum for an annotated span (the majority in this case) is most valuable. A negative phenomenon is a disunity of opinions, while two matching opinions are arguable, but worthy of attention as coinciding.

In a group of 3 volunteers a quorum for three is most valuable as it is the general view which wouldn't change its character of quorum after the 4th volunteer would annotate the same data. On the other hand, a matching opinion of two can either become a quorum (3 of 4), or get an opinion of 2 out of 4 volunteers; both cases are mentioned above.

In a group of two the markup is in an initial state. 2 matching opinions are a positive phenomenon, but it can change significantly (this would be discussed below).

The markup data are represented as Venn diagrams for 2, 3, 4 volunteers, i.e. diagrams for each combination of volunteers who appeared to collaborate within at least one paragraph. A diagram consists of intersecting Euler rings, each comprising opinions of one volunteer from the group described in this diagram – the team that examined together a set of paragraphs. Cases when one set of paragraphs is examined by two volunteers, and the other one – by four including the first two, are considered as completely different and are described in two distinct diagrams (for 2 and for 4 volunteers), which do not intersect in any way.

Annotated data of a corpus are represented as three classes of data: paragraphs examined by only two, only three, only four volunteers. An attempt at joining the set of Venn diagrams of power 2 (which characterize opinions of pairs of volunteers) with all existing pairs which can be found within groups of four or three, would result in considering partial, inconsistent data: in full data, the quorum of four has a different degree of confidence, a higher one, while 2 matching opinions among 2 volunteers are an intermediate stage of markup which may change significantly. Thus, duos should be interpreted separately from the bigger groups. In this research, an aggregate picture is considered for the corpus: paragraphs are grouped regarding sizes of groups which annotated those paragraphs (equal sizes are the classifying criterion).

Hence all of the paragraphs examined by the same team are included in one diagram. Next, indexes of agreement should be calculated per each diagram.

Each paragraph p_i is annotated by M_i volunteers: $M_i = |\{u_j: u_j \text{ annotated } p_i\}|$, $M_i \in [1, \Omega]$. $\Omega = 5$: OpenCorpora's standart $\Omega = 4$ plus moderator annotation. There are Q_i opinions on a paragraph p_i : $Q_i = |\{s_j: s_j \in p_i\}|$ is a number of spans annotated by volunteers in p_i (spans annotated by different volunteers can match if opinions do).

For p_i a Venn diagram is constructed, which comprises opinions (spans) by M_i volunteers who annotated p_i . M_i is the power of this diagram, which comprises all of

the spans annotated by a given group of volunteers. Basing on sets of intersecting and non-intersecting opinions (spans) we introduce following criteria:

- (1) number of equal opinions of α of M_i volunteers, including the given volunteer ($2 \leq \alpha \leq M_i$), on the presence of a given span;
- (2) number of standalone opinions, shared by no one else.

From all of the Venn diagrams with power M_j , comprising a given volunteer V , statistics can be obtained for this volunteer, namely numbers of observed cases of agreement and disagreement with V : V annotated a span, this opinion is shared by α volunteers, including V ($1 \leq \alpha \leq M_j$, $1 < M_j \leq 4$, j is the index of a given diagram in a set of all diagrams comprising markup by the volunteer V). These would be criteria of agreement of 1/2 (one of two), 2/2, 1/3, 2/3, 3/3, 1/4, 2/4, 3/4, 4/4 volunteers with V (including V). Each criterion is a sum of similarly named criteria from diagrams comprising V 's annotations.

Statistics over a corpus is a set of statistics over all of the volunteers. Each volunteer is characterized with the agreement data in groups of 2, 3, 4 volunteers, where one conducted annotation. Cumulative statistics over a corpus is formed with sum of criteria over all volunteers— cumulative criteria of agreement of N of M volunteers (marked as N/M). 4 categories of agreement criteria are pointed out; each category is of different value.

Category I – quorum when matching majority of opinions (3 opinions or more): 4/4, 3/4, 3/3 volunteers. Estimated as a positive phenomenon: most volunteers got similar opinions, and these are data which wouldn't change category (in case 3/3 the appearance of the 4th volunteer wouldn't change the quorum state).

Category II – possible quorum, shared opinions in incomplete groups of two and three volunteers (2/2, 2/3), which would likely get to quorum during further annotation by upcoming volunteers. Estimated as a positive phenomenon.

Category III – opinions divided. Standalone opinions among two annotators (1/2) which could become quorum during further annotation, as well as opinions shared by a half of four volunteers (2/4). Estimated as a controversial point; part of opinions could change category during further annotation.

Category IV – minority of opinions, not shared by the majority and with no chance to change category to majority: 1/3, 1/4 volunteers won't get to the first two categories. Estimated as negative phenomenon.

The introduced approach is based on estimating the contribution of each category of criteria. The more the contribution of positive phenomena (the first two categories), the higher the agreement. The four categories should be represented as a histogram, using color notation. The same data should be organized into a pie chart in order to compile the four categories of criteria.

As a major goal of this work is to select a single criterion for overall annotation estimation (for a corpus snapshot), the sum of two categories of criteria corresponding to positive phenomena would be such **total criterion**.

An illustration of replenished annotation is given in categories on Fig. 2.

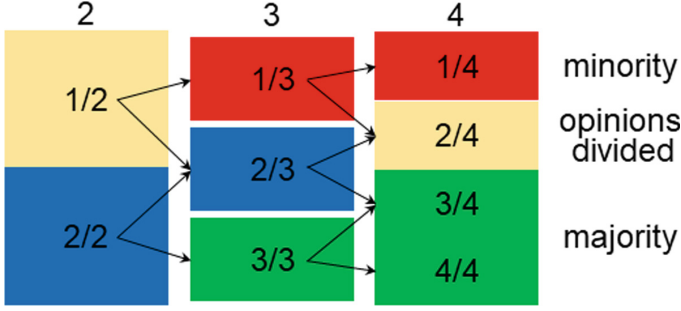


Fig. 2. Categories of agreement criteria

In all, the suggested method for IAA evaluation for the NEA task is as follows.

- (1) Constructing Venn diagrams for each of existing groups of volunteers: calculating the number of cases of agreement of α annotators on the presence of each unique span in paragraphs annotated by the given group.
- (2) Calculating statistics for each volunteer, namely cases of shared opinions from each Venn diagram comprising this volunteer's annotations (agreement of 1/2, 2/2, 1/3, 2/3, 3/3, 1/4, 2/4, 3/4, 4/4 annotators, including the one under review).
- (3) The cumulative statistics should be obtained for the corpus: criteria of the same name should be aggregated over all of the volunteers (agreement of 1/2, 2/2, 1/3, 2/3, 3/3, 1/4, 2/4, 3/4, 4/4 annotators).
- (4) The contribution of each of the above mentioned four categories of agreement criteria should be estimated, resulting in one final criterion.

4 Approbation

4.1 Setting

The IAA evaluation is given for corpus data without moderator's markup, as the task is to evaluate the agreement before moderation. In OpenCorpora a motivated community conducts annotation without any form of reward. But as the time was limited, at a certain stage the corpus comprised markup by assigned volunteers, motivated by the employer (they received an assignment to annotate named entities). The amount of annotated data which served for approbation is given in Table 1. Subcorpora are highlighted, each containing groups of attributes describing a single type of entity. The evaluation is given for these subcorpora as well.

In Sect. 4.5 the IAA will be evaluated for an alternative experiment on a NEA task over a recipes corpus. 12 attributes should be annotated: food, value, unit, process, action, tool, tool modifier, action modifier, action result, state adjective, result verb, measure.

Table 1. Amount of spans within groups of tags.

Corpus	Full data	Person tags	Job tag	Location tags	Organization tags
Full	71667	23356	7222	18852	22237
Motivated volunteers	40667	12980	4196	10399	13092

4.2 Evaluation of IAA for Full Corpus

The evaluation of the agreement in the final corpus is shown on Fig. 3. The histogram on Fig. 3(A) shows cumulative indexes of agreement (comprising corresponding opinions), colored according to categories: I – green, II – blue, III – yellow, IV – red (the latter is shown below the axis as it is considered to be a negative phenomenon). The pie chart on Fig. 3(B) shows compiled criteria of agreement as in four categories. Figure 3(C) shows the amount of paragraphs and parts of paragraphs annotated by volunteer groups of different size. The cumulative investment of positive agreement phenomena (green

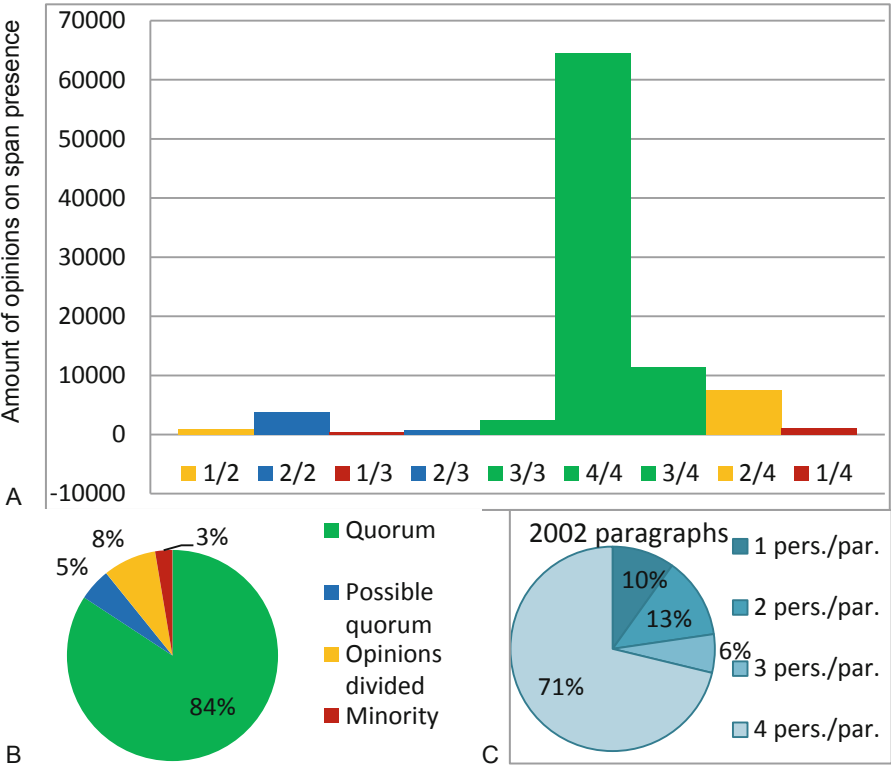


Fig. 3. IAA evaluation by categories – full corpus (Color figure online)

and blue zones) and of negative phenomena (the red zones) should be estimated basing on data shown on Fig. 3(A) and (B).

Quorum category value is 84%, possible quorum – 5%, in the aggregate these positive phenomena prevail and constitute 89%. This value exceeds the IAA threshold of 80%, stated in [1]. The minority category constitutes 3%. The conclusion is, the volunteers are shown to agree consistently.

At early stages of markup (2% to 15% of full corpus data, no additionally motivated annotators involved at these stages) quorum and possible quorum values are over 83%. Thus, at early stages of markup annotators agreed consistently as well. Obtaining IAA statistics for early-stage markup (before moderation) was the goal of this work; the introduced approach allowed achieving this goal.

4.3 An Experiment with Motivated Volunteers

Following the OpenCorpora tradition, the markup is made by volunteers. But since a big amount of data was required for a contest, and the amount was inversely proportional to the time available, an experiment was conducted on involving motivated annotators, with the support of ABBYY.

IAA arises insignificantly (by 1–2%) for a subcorpus comprising markup by annotators to whom the NEA task was imposed as job to do (see Table 1). This is the evidence to the accuracy of these annotators, as well as of all of the volunteers. Thus, the hypothetic transition from crowdsourcing to rewarded crowdsourcing in the NEA task doesn't show significant change of agreement (but a huge amount of high-quality markup). All of the volunteers of OpenCorpora's are accurate and do agree in general.

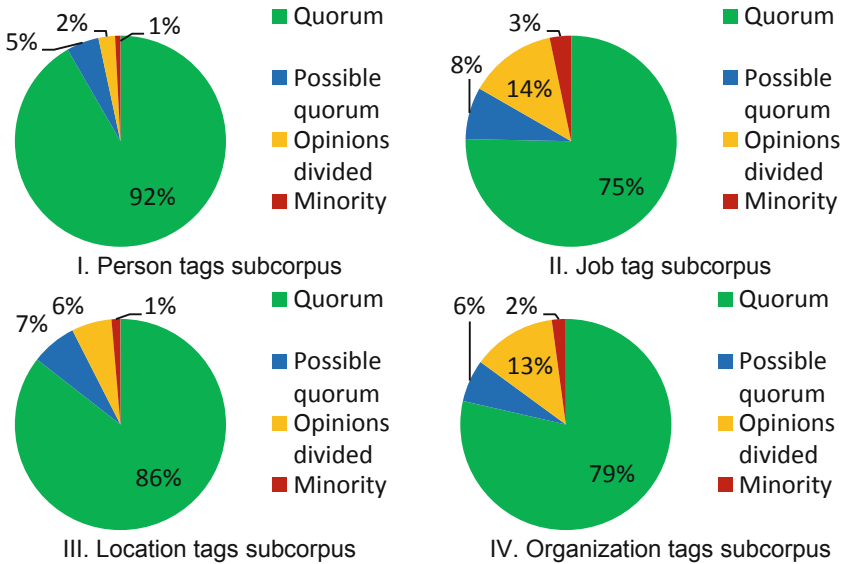


Fig. 4. IAA evaluation by categories – subcorpora

4.4 Detailed Evaluation of IAA for Subcorpora by Tags Groups

The evaluation of IAA was also conducted for subcorpora comprising groups of tags (see p. 4.1). The results of such detailed evaluation over parts of the full corpus are shown on Fig. 4. Positive category values vary from 83% to 97%, hence, annotators tend to agree. The highest value of 4/4 quorum is achieved on the person tags subcorpus (84%), which testifies that person tags are annotated most uniformly.

4.5 Evaluation of IAA for a Corpus of Recipes

IAA evaluation was also conducted for an alternative NEA corpus of recipes attributes, as the conceptual model described above (see Sect. 1.1) is abstract and can be used for different annotation tasks (defining its own set of spans types). The results are shown on Fig. 5. The results are not that highly rated: quorum and possible quorum constitute 18% and 54% respectively, which amounted to 72%. This latter value doesn't exceed the threshold of 80% established in [1]. The results are not so good, which may result from the nature of the data (a large number of attributes should be annotated – 12 items, see p. 4.1) and infers the need either for clarifying the guidelines for annotators, or for examining the data unambiguity.

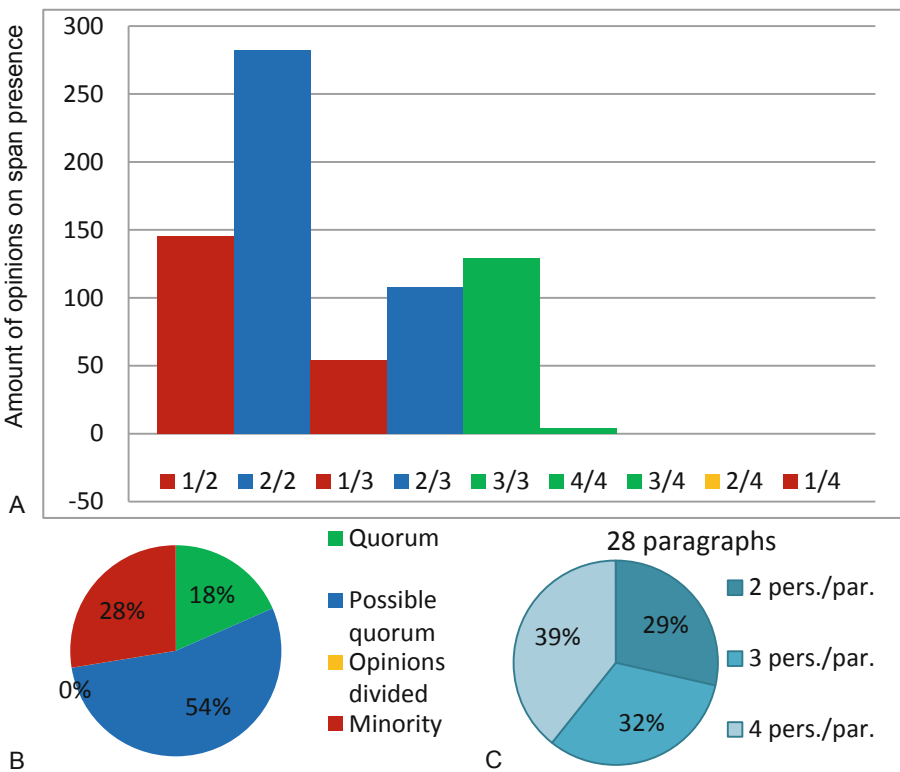


Fig. 5. IAA evaluation by categories – recipes corpus

5 Conclusion

A procedure is suggested for inter-annotator agreement estimation for the named entities annotation task, based on selecting an open set of chains of tokens from a paragraph. This approach overcomes the constraint of applicability of existing methods, and allows automated estimation of IAA for the NEA task.

An approbation of the suggested method was conducted basing on named entities, annotated in the crowdsourcing project OpenCorpora. The calculated total agreement value of 87% of positive cases is the evidence to the success of the markup initiative, along with agreement values over 83% at early stages of markup. This testifies that annotators agree in general, which supports the fundamental assumption that they annotate according to the guidelines and share the same understanding of it.

The introduced method can be applied to IAA evaluation for the NEA task at early stages of markup, before moderation or validation. Several prospects exist for further development of this research, particularly, developing indicators for evaluating separate volunteers' statistics, sensitive to whether one's opinions get most often in categories of quorum or minority.

This approach to evaluating inter-annotator agreement in data with open set of alternatives is also applicable in tasks where an annotator should select characteristic features from a text, e.g. for functional style annotation in order to select the markers (words or structures) corresponding either to one or several functional styles, or to a combination of styles (as writers as E.M. Remark work as publicists, and hence their style is rather a publicistic one when whitening novels).

One more application is annotating multimodal communicative behavior, as in Russian Emotional Corpus (REC). REC comprises videos which are subject to annotate on different layers: speech, gestures, gazes and narrower layers (e.g. gestures with hands, with head) [9]. Annotations might intersect as well as in the title task because behavior elements can occur simultaneously. The application of such corpora is in developing a set of multimodal reactions for robot-companions to simulate communicative behavior in order to make robot-companions look more natural to human interlocutors [8].

Acknowledgements. First, authors wish to thank all of the volunteers of the OpenCorpora project. Second, authors are grateful to Olga Plotnikova and Nikita Golovanyov for their contribution into laying out the visualization. Last, but not least, we would thank Olga Makarova for priceless remarks, which helped to complete the general idea of this research.

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